

Theoretical Astrophysics: Stellar Structure and Evolution

Homework 4

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Deadline for this homework is 31/08 23:59

Important: Please do not turn in jupyter notebook, but rather a textual report, in pdf format or similar.

Problem 1: Show that, for a blackbody radiation, radiation pressure is equal to:

$$p_{\text{rad}} = \frac{u}{3} = \frac{4\sigma}{3c}T^4. \quad (1)$$

Hint: One of the ways to solve this is to consider that the radiation is impinging on a perfectly absorbing wall and that the intensity of the radiation is given by the Planck function (recall the first homework). Then relate the intensity of the radiation to the number of photons, and that number of photons to the momentum the photons are transferring to the wall.

Problem 2: Following sketches from the lecture on helioseismology, derive the equation for the soundspeed, i.e. for the so called “p-modes”.

Problem 3: Add gravity to the system that was used to demonstrate sound waves on the lectures and in the previous problem and derive the equations governing internal gravity waves or “g-modes”.

Problem 4 (Bonus problem!!!): Run a MESA model for a desired stellar mass and metallicity: <http://user.astro.wisc.edu/~townsend/static.php?ref=mesa-web-submit>

And visualize the evolution of the fundamental stellar parameters with time (mass, radius, luminosity).

Then, plot and discuss stellar structure for a chosen moment from stellar life on the main sequence.

Useful physical constants

- $R_{\odot} = 696 \times 10^6 \text{ m}$
- $M_{\odot} = 1.989 \times 10^{30} \text{ kg}$
- $L_{\odot} = 3.83 \times 10^{26} \text{ W}$

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- $T_{\odot}^{\text{eff}} = 5777 \text{ K}$
 - $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$
 - $c = 2.997 \times 10^8 \text{ m/s}$
 - $G = 6.674 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
 - $k = 1.38 \cdot 10^{-23} \text{ J/K}$
 - $m_{\text{H}} = 1.67 \cdot 10^{-27} \text{ kg}$
 - $h = 6.626 \times 10^{-34} \text{ J s.}$
 - $k = 1.38 \times 10^{-23} \text{ J/K.}$
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